

Business Problem & Project Objective

Companies often leverage coupons as a marketing strategy to attract new customers, retain existing ones, and increase customer spending. However, coupon distribution can be costly, and its effectiveness heavily relies on whether potential customers would accept and use them. To maximize the return on investment (ROI) in coupon marketing, it's critical for a business to predict which customers are more likely to accept a coupon under various scenarios.

This project aims to develop a Deep Neural Network (DNN) model that can accurately predict whether a person will accept a coupon under various scenarios and personal attributes.

By successfully achieving this, the model will provide valuable insights to guide the business's coupon distribution strategy, ultimately optimizing marketing spend and increasing overall ROI. It will help the business target customers who are more likely to accept and use the coupons, leading to enhanced customer engagement and sales.

Dataset

The dataset is available in UCI Machine Learning Repository. This dataset, collected via an Amazon Mechanical Turk survey in 2017, includes 12k instances and encompasses 25 original categorical features. This covers a range of factors that could potentially affect an individual's decision to accept a coupon. The factors include:

- **Scenario factors:** Destination, current time, weather, the type of coupon, coupon expiration time, and the passengers in the car.
- **Personal attributes:** Gender, age, marital status, presence of children, level of education, occupation, and income.
- **Activity frequency:** How often the individual visits a bar, a coffeehouse, or a restaurant. The frequency is considered for two types of restaurants - one with an average expense per person of less than \$20 and the other with an average expense of \$20 to \$50.
- **Coupon-related factors:** Whether the driving distance to the place (restaurant/bar) where the coupon could be used is more than 15 or 25 minutes, and whether the place is in the same direction as the individual's current destination.

The outcome is a binary variable, indicating whether the coupon was accepted or not, and is fairly balanced, with 6877 instances (57%) labeled '1' (coupon accepted) and 5202 instances (43%) labeled '0' (coupon not accepted). The data allocation for modeling is to split the data into a training dataset (70%, 8.4k instances), a validation dataset (20%, 2.4k instances), and a test dataset (10%, 1.2k instances).

Data Processing

Our data processing procedure consists of three key steps, aimed at cleaning and transforming the dataset to make it suitable for our model:

1. Missing Value Handling: The "car" feature, which nearly only contained NA values (12,576 NA values out of 12,684 total instances), was removed. Then, we identified a set of 794 data entries with missing values across five features: "Bar", "CoffeeHouse", "CarryAway", "RestaurantLessThan20", and "Restaurant20To50". Instead of discarding these five features, we opted to remove only these 794 data entries.

2. Eliminate Feature with Single Value: we discarded the feature “toCoupon_GEQ5min” because it included only a single value, “1”, providing no variance and hence no valuable information for our model.

3. Categorical Variables Conversion: Since our dataset is survey data with categorical variables, these categorical variables required conversion to either numerical variables or dummy variables for the further modeling process.

- **Numerical Conversion:** 8 features that held intuitive numerical meaning were converted to numerical variables. For instance, if a person has never been to a bar, the category “never” was converted to the number “0”.

```
Variables below were converted from categorical to numerical variables:  
  
'temperature', 'age', 'income', 'Bar', 'CoffeeHouse', 'CarryAway',  
'RestaurantLessThan20', 'Restaurant20To50'
```

The code below shows the mapping between categories and corresponding numbers:

```
temperature_value_map = {'30':30, '55':55, '80':80}  
age_value_map = {'below21': 21, '21':21, '26':26, '31':31, '36':36, '41':41, '46':46, '50plus': 50}  
income_value_map = {'Less than $12500':12500,  
                    '$12500 - $24999':18750,  
                    '$25000 - $37499':31250,  
                    '$37500 - $49999':43750,  
                    '$50000 - $62499':56250,  
                    '$62500 - $74999':68750,  
                    '$75000 - $87499':81250,  
                    '$87500 - $99999':93750,  
                    '$100000 or More':100000}  
Bar_value_map = {'never':0, 'less1':1, '1-3':2, '4-8':6, 'gt8':8}  
CoffeeHouse_value_map = {'never':0, 'less1':1, '1-3':2, '4-8':6, 'gt8':8}  
CarryAway_value_map = {'never':0, 'less1':1, '1-3':2, '4-8':6, 'gt8':8}  
RestaurantLessThan20_value_map = {'never':0, 'less1':1, '1-3':2, '4-8':6, 'gt8':8}  
Restaurant20To50_value_map = {'never':0, 'less1':1, '1-3':2, '4-8':6, 'gt8':8}
```

- **Dummy Variables Creation:** The remaining 15 categorical variables were converted into dummy variables to simplify their representation.

```
Variables below were converted from categorical to dummy variables:  
  
'destination', 'passanger', 'weather', 'time', 'coupon', 'expiration', 'gender',  
'maritalStatus', 'has_children', 'education', 'occupation', 'toCoupon_GEQ15min',  
'toCoupon_GEQ25min', 'direction_same', 'direction_opp'
```

4. Eliminate Perfectly Correlated Features: In the final step, we examined the correlation matrix of our dataset and found that the feature “time_7AM” was perfectly correlated (correlation of 1) with “destination_Work” and the feature “direction_opp_1” was perfectly inversely correlated (correlation of -1) with “direction_same_1”. These correlations indicate redundancy as one feature can be linearly predicted from the other. Therefore, we removed one feature from each of these two pairs, specifically “destination_Work” and “direction_opp_1”, to eliminate this redundancy.

By the end of the data processing stage, we had a refined dataset with a total of 61 features, we then did **data normalization** for numerical features before the subsequent feature selection step.

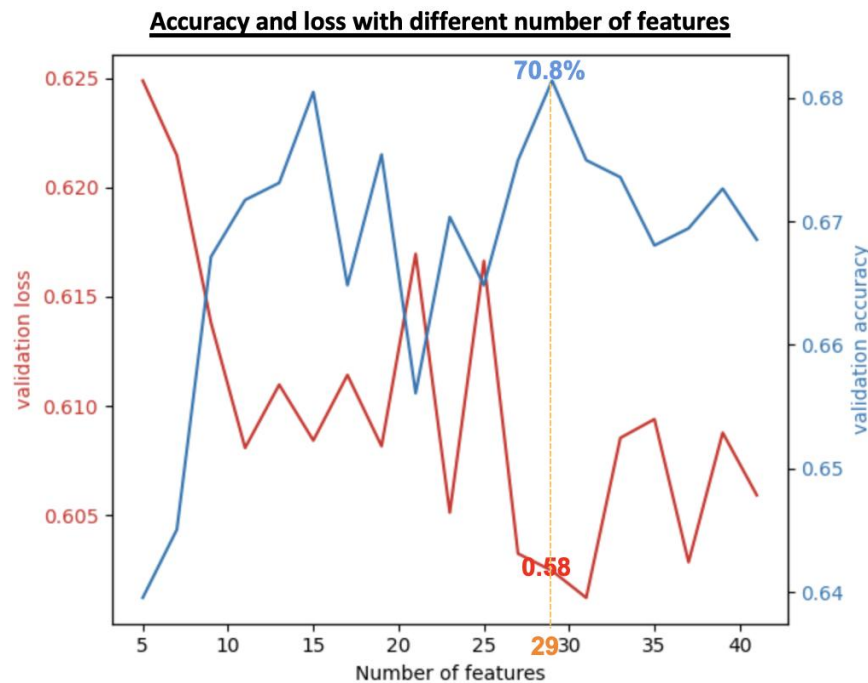
Feature Selection

The feature selection process for this project consists of two main parts:

- **Identifying Feature Importance:** The process starts by tuning an XGBoost Classifier in the training set to evaluate feature importance. Here, feature importance is determined based on the impact each feature has on the model's predictive accuracy. In this stage, we found that 20 out of the total features had zero importance, i.e.,

they contributed nothing towards the model's predictive ability. These non-essential features were subsequently dropped, resulting in a refined set of 41 valuable features that were retained for further utilization.

- **Determining the Optimal Number of Features:** In the second part of the feature selection process, we developed a baseline model and experimented with different numbers of features within this model. The feature examination was carried out according to their importance rankings. In this approach, we start with a small group of 5 features and add 2 more at a time until we've tried all 41 features. The optimal number of features - the count that yielded the highest validation accuracy - was selected for further modeling. As shown in the figure below, the optimal number of features is 29, which brings the highest validation accuracy on the baseline model.



Model Structure

Best model

	structure	Test Loss	Test Accuracy
Dropout model	DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	0.61	73.8%

Experiments

Experiments Overview

This is baseline model: DNN(B)-frs(5240)-[12(relu)-8(relu)-1(sigmoid)]-[ep(30)]. We do experiments from this model.

We used a systematic approach to develop and optimize a deep neural network (DNN) model for a machine learning task.

1. First, we conducted an initial experiment to adjust the number of layers and neurons in a large range, and selected the model with the highest accuracy and lowest loss as the baseline model.

2. Next, we performed further experiments to fine-tune the model's architecture by testing various numbers of neurons in the first and third hidden layers for different batch sizes. We also tested the top-performing models for different epochs to determine the optimal number of epochs for our task.

Experiment Details

The baseline DNN model was tailored to the optimal feature count established in the preceding feature selection phase. The experimental plan aimed at enhancing model accuracy and efficiency, while ensuring robustness against overfitting and improved generalization.

- 1. **Architectural Refinements:** Our primary focus was adjusting the number of layers and neurons extensively. This allowed us to compare their performances and designate the highest accuracy, lowest loss model as our baseline.
- 2. **Hidden Layer and Batch Size Adjustments:** We paid particular attention to the baseline model's hidden layer architecture. We experimented with different neuron counts in the first and third hidden layers, assessing their performance for batch sizes of 20 or 40.
- 3. **Epoch Optimization:** The top-performing models, in terms of loss and accuracy, underwent further testing for different epochs. This aimed to determine the optimal **number of epochs for our task**.
- 4. **Overfitting Mitigation and Generalization Improvement:** To prevent overfitting and enhance the model's generalization ability, we incorporated L1 regularization and Dropout into our models for comparison with the baseline.
- 5. **Model Refinement and Testing:** The model performance was continually monitored, with the most optimal model selected for additional refinement and testing.

Performance

Among our current experiments(71 in total), 3 DNN models that achieved the highest test accuracy are shown as below:

layer	epochs	batch_size	base: Loss	base: Accuracy	Best result	L1: Loss	L1: Accuracy	Best result	dropout: Loss	dropout: Accuracy	Best result
DNN(B)-frs(5240)-[105(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.142	73.18%	best result	0.614	70.78%		0.614	71.61%	
DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.300	70.94%		0.614	70.86%		0.608	73.76%	best result
DNN(B)-frs(5240)-[30(relu)-12(relu)-24(relu)-12(relu)-8(relu)-1(sigmoid)]-[ep(30)]	30	40	0.593	69.54%		0.612	72.02%	best result	0.689	57.70%	

Conclusion

Our final model demonstrated a satisfactory performance, achieving a 73.8% accuracy on the test data. This model effectively incorporated 29 key features, spanning diverse categories such as Scenario Factors, Personal Attributes, Activity Frequency, and Coupon-Related Factors.

The optimized model facilitates the prediction of customer responses to coupons, providing valuable insights for businesses. By accurately identifying potential customers who are more likely to accept coupons, businesses can target their marketing efforts more effectively. This, in turn, is expected to boost customer engagement and drive sales, affirming the practical value and potential of our model in a real-world business context.